

ARTIFICIAL INTELLIGENCEAnalytical problem Solving1. A search problem

Node	Heuristic value $h(n)$
A	10
B	8
C	5
G	0

cost from $A \rightarrow B = 3$

cost from $A \rightarrow C = 4$

formula:

$$f(n) = g(n) + h(n)$$

where:

* $g(n)$ = Actual cost from start node

* $h(n)$ = Heuristic value

* $f(n)$ = Estimated total cost

calculation:

for Node B:

$$g(B) = 3$$

$$h(B) = 8$$

$$f(B) = 3 + 8$$

$$= 11$$

1. A search problem

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* $f(n)$ = Estimated total cost

calculation:

for Node B:

$$g(B) = 3$$

$$h(B) = 8$$

$$f(B) = 3 + 8$$

$$= 11$$

for Node c:

$$g(c) = 4$$

$$h(c) = 5$$

$$f(c) = 4 + 5 \\ = 9$$

Result

Node	$g(n)$	$h(n)$	$f(n)$
B	3	8	11
C	4	5	9

Conclusion:

Since Node c has the smaller $f(n)$ value (9),

* Search expands Node c first.

2. A Search - Google Maps Example*

Given

Route	Distance travelled $g(n)$	Estimated Rem Distance $h(n)$
Route 1	40 km	80 km
Route 2	50 km	60 km

Formula:

$$f(n) = g(n) + h(n)$$

calculation:

Route 1:

$$f(n) = 40 + 80$$

$$\therefore f(n) = 120 \text{ km}$$

Route 2

$$f(n) = 50 + 60$$

$$f(n) = 110 \text{ km}$$

Result:

Route	$g(n)$	$h(n)$	$f(n)$
Route 1	40	80	120 km
Route 2	50	60	110 km

Conclusion:

Since Route 2 has the smaller $f(n)$ value (110 km), A* Search selects Route 2 as the preferred route.

3. Hill climbing Algorithm

Given

Current State value = 15

Neighbouring state values: 12, 18, 20, 17

Working:

Hill climbing selects the neighboring state with the highest value (best improvement)

Among the neighbors:

* 12

* 18

* 20 ✓

* 17

Result:

The algorithm moves to the state with value 20

Conclusion:

Since 20 is the highest neighboring value, the hill climbing algorithm chooses state 20 because it provides the maximum immediate improvement toward the goal.